Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad — July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering — *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi — Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering — *CGPA: 9.05*

Research & Open-Source Experience

Core Contributor, RookDB — Aug 2026 – Present
Independent Study, DSAC Lab, IIIT Hyderabad | Rust, SQL, Databases — *Hyderabad, India*

- Co-designed the lab's open-source SQL database in Rust: storage engine, buffer pool, B+Tree indexes and query executor.
- Built the cost-based query planner and `EXPLAIN ANALYZE`, and used them to find and fix slow queries.
- Led performance work that made the benchmark suite **14x faster**, with **1,200+ tests** including comparison with SQLite.

Projects

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

IndicLLM: Transformer Pretraining & Reasoning SFT | *Python, PyTorch, SentencePiece, ONNX*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Tested whether the models were reasoning or memorising answers before reporting results.

Warehouse Traffic Control: Deterministic Distributed Simulator | *JavaScript, Web Workers, Three.js*

- Built a reproducible simulator of a 200-robot warehouse where all components talk over an unreliable network.
- Designed a lease-based protocol with **zero collisions** over 416K moves, even with lost messages, clock drift and crashes.
- Ran parameter sweeps in parallel, and found that longer runs overturned conclusions drawn from short ones.

Technical Skills

Languages: C/C++, Python, Rust, SQL, Bash, JavaScript.
Machine Learning: PyTorch, NumPy, Pandas, Scikit-learn, ONNX.
Systems & Tools: Linux, Multithreading, Sockets, Git, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.